Project Proposal

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October 2026

We want to build a JavaScript runtime for react-reconciler compatible renderers (react-native, my renderer, ink). The vision for this would be emulating some version of a browser (combining the graphics layer + the pseudo-OS layer that most browsers have ended up in). Our focus is on providing a useful interaction layer with the OS while also providing some isolation. This is similar to what Node, Deno and Bun do (take existing runtime and add support for OS interaction and functionality standalone from the browser), but with a smaller subset of functionality.

We want to provide the runtime with access to the file system, FFI, some syscalls (exec, fork), and if we have time, a tab/process abstraction for true multiprogramming.

For the JS runtime, we plan on starting from JSCore and building out the sandboxed environment that a JS developer would work with. While JSCore is a C++/Swift/Objective-C based runtime, our implementation of the sandboxed OS interaction layer will be in Rust. By doing this, we hope to isolate the JavaScript runtime from unauthorized access to the OS.

We plan on verifying our runtime by building out a semi-formal specification in comments and documentation, then using these to drive how we write invariants (pre and post conditions) in our functions. We will also write a test suite for this library, but this probably won't guarantee full isolation.